

5.5.3 Cosmology

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) distances measured in astronomical unit (AU), light-year (ly) and parsec (pc)	M4.6
(b) stellar parallax; distances the parsec (pc)	
(c) the equation $p = \frac{1}{d}$, where p is the parallax in seconds of arc and d is the distance in parsec	
(d) the Cosmological principle; universe is homogeneous and the laws of physics are universal	
(e) Doppler effect; Doppler shift of electromagnetic radiation	
(f) Doppler equation $\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ for a source of electromagnetic radiation moving relative to an observer	
(g) Hubble's law; $v \approx H_0 d$ for receding galaxies, where H_0 is the Hubble constant	
(h) model of an expanding universe supported by galactic red shift	
(i) Hubble constant H_0 in both $\text{km s}^{-1}\text{Mpc}^{-1}$ and S^{-1} units	
(j) the Big Bang theory	

- (k) experimental evidence for the Big Bang theory from microwave background radiation at a temperature of 2.7 K HSW7, HSW11 The development and acceptance of Big Bang theory by the scientific community.
- (l) the idea that the Big Bang gave rise to the expansion of space-time
- (m) estimation for the age of the universe; *M1.4*
 $t \approx H_0^{-1}$
- (n) evolution of the universe after the Big Bang to the present
- (o) current ideas; universe is made up of dark energy, dark matter, and a small percentage of ordinary matter.

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